

AI DRIVEN BLOOD TEST DISEASE PREDICTION SYSTEM

Artificial Intelligence & Data Science

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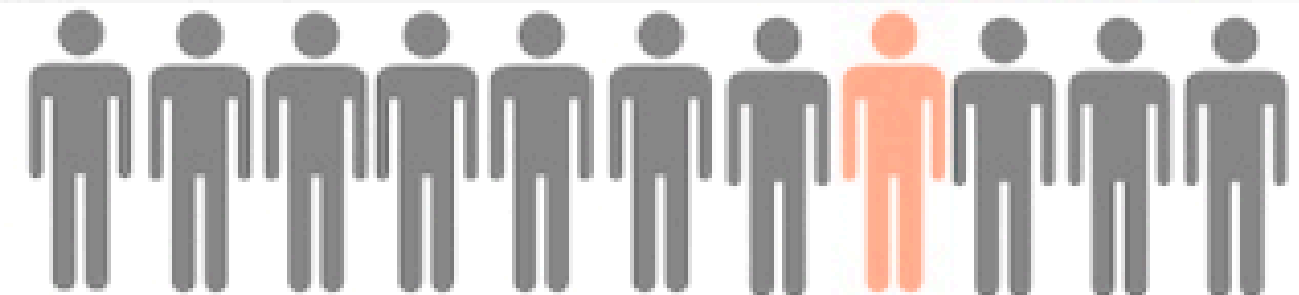
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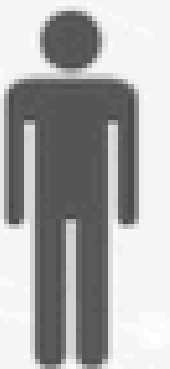
INTRODUCTION

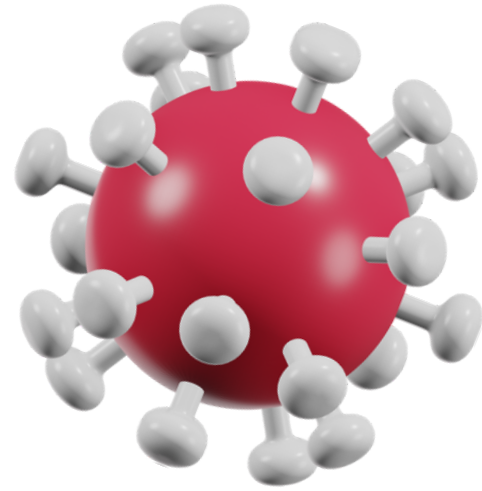
- The project focuses on predicting diabetes, as **Diabetes** is a chronic medical condition that affects millions of people worldwide.
- **1 in 2** adults with diabetes are **undiagnosed** as they don't even know that they are suffering from an untreatable disease
- The goal of our project is to develop a **Blood Test Disease Prediction System** that utilizes machine learning techniques to analyze blood test data and accurately predict whether a patient is diabetic or non-diabetic.
- This approach supports early detection, personalized care, and better healthcare outcomes through scalable, real-time risk assessment systems.

1 in 11 adults has diabetes (**425 million**)



1 in 2 adults with diabetes is undiagnosed (**212 million**)





PROBLEM STATEMENT

- Traditional blood test analysis requires manual interpretation, leading to delays in detecting diseases.
- Many regions lack access to advanced labs, making timely disease detection difficult.
- Blood test reports contain diverse formats and unstructured data, making automation challenging.
- Existing systems lack AI-driven solutions to process and predict disease risks efficiently.



OBJECTIVES

- Our aim is to develop an AI-powered system to analyze blood test data for early disease detection.
- Our motive is to provide a cost-effective and scalable solution for remote and underserved healthcare regions.
- To assist healthcare providers with early detection and risk assessment.
- To provide fast, cost-effective, and data-driven clinical decision support for patients and doctors.



LITERATURE REVIEW

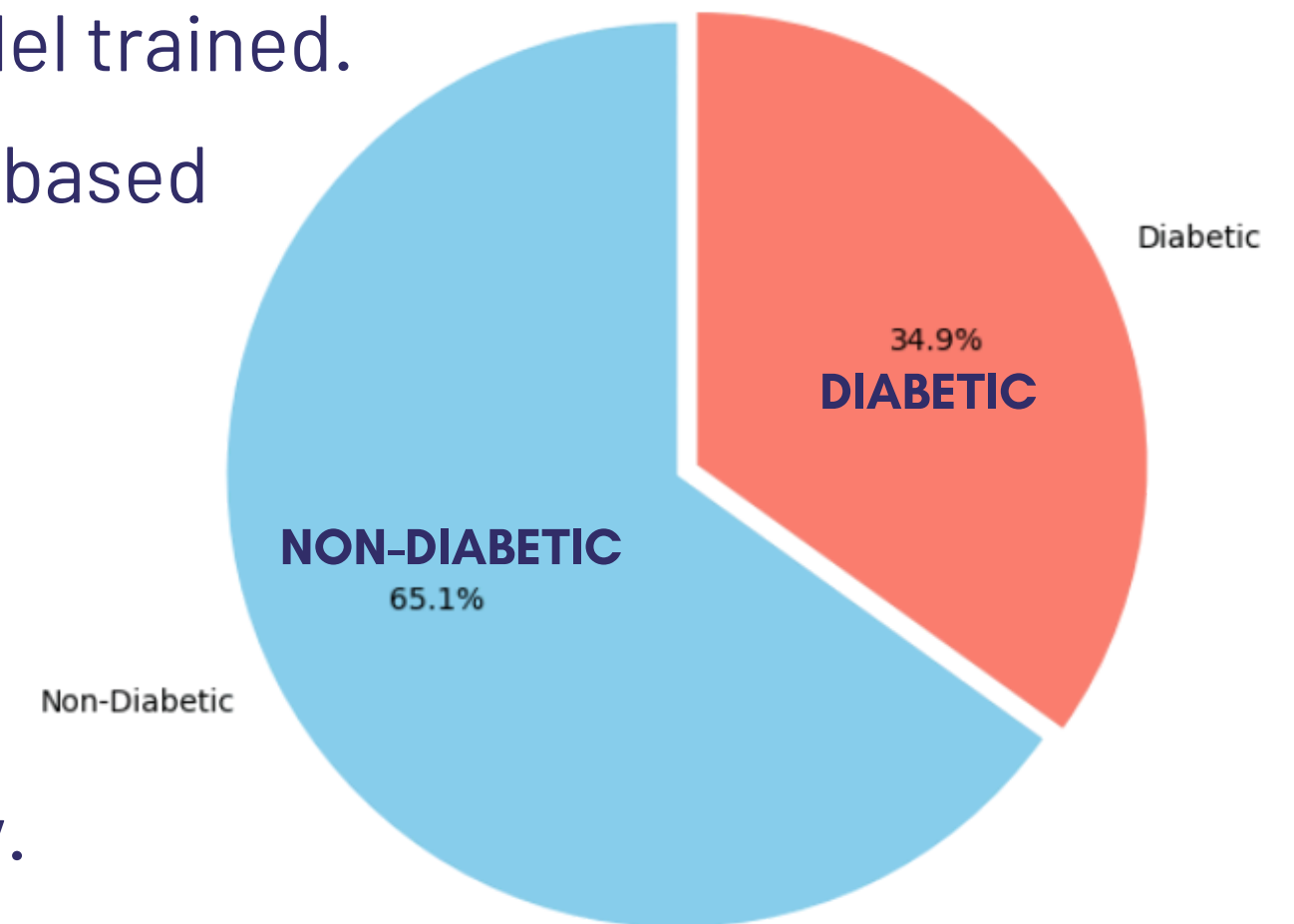
- **Sneha & Gangil (2019)** optimized feature selection, achieving 98% specificity using Decision Trees and Random Forests.
- **Ahmed et al.** developed a hybrid SVM-ANN model, attaining 94.87% accuracy.
- **Laila et al.** validated ensemble learning techniques, with Random Forest reaching 97% accuracy.
- **Wadekar et al.** introduced mobile AI-based health monitoring.
- **Padhy et al.** integrated IoT-driven real-time disease tracking, achieving 98.4% accuracy.

AI-driven solutions improve predictive accuracy, scalability, and accessibility, advancing automated disease detection.



METHODOLOGY

1. **Data Collection:** Gather historical medical datasets.
2. **Data Preprocessing:** Clean and normalize data by handling missing values
3. **Model Selection:** Choose appropriate ML models (e.g., logistic regression, SVM) based on data type and problem.
4. **Model Training:** Train the selected models on a training dataset.
5. **Model Evaluation:** Check the accuracy and Precision of the model trained.
6. **Prediction:** Use the trained model to predict disease outcomes based on new blood test data.
7. **Deployment:** Deploy the model in a user-friendly interface for real-time predictions.
8. **Monitoring and Maintenance:** Continuously monitor the model's performance and update with new data when necessary.



Work Flow



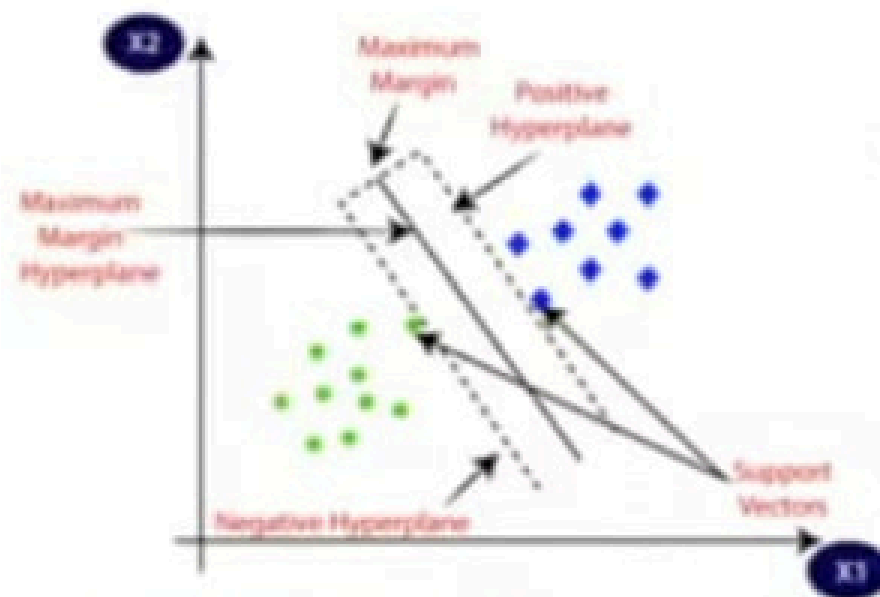
Diabetes Data



Data pre processing



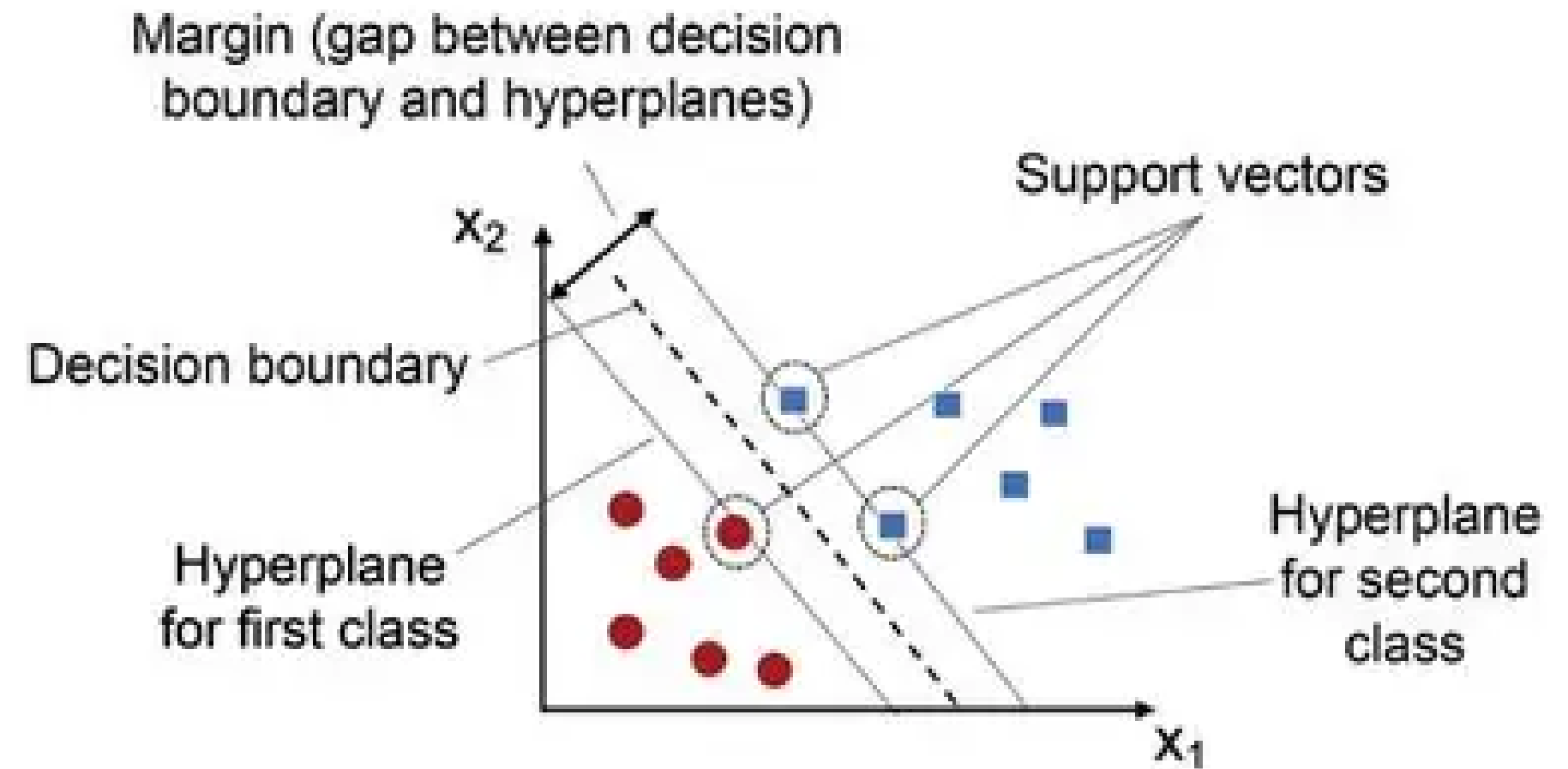
Train Test split



Support Vector Machine Classifier

SUPPORT VECTOR MACHINE

- SVM is a **supervised machine learning algorithm** used to classify data into categories.
- It finds the best possible boundary (**hyperplane**) that separates different groups.
- **Ensures maximum separation** between classes for better generalization.
- Support vectors are the closest points to this boundary and help define it.
- It is widely used in **medical diagnosis**, **text classification**, and **image recognition** due to its accuracy.



HARDWARE & SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

- **Processor:** Intel i3 or equivalent.
- **RAM:** 4 GB minimum.
- **Storage:** 250 GB free hard disk space.
- **Graphics Card:** Not required.
- **Internet:** Required for downloading libraries and updates.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10 or any Linux distribution.
- **Programming Language:** Python (version 3.6 or above).
- **Integrated Development Environment (IDE):** Jupyter Notebook or any text editor.
- **Libraries:** Pandas, NumPy, scikit-learn.



FUTURE SCOPE & CONCLUSION

- Integration with Other Medical Data like ECG, medical history for more accurate predictions.
- Implement real-time disease prediction using live blood test data.
- Enhance models with advanced techniques like deep learning for better accuracy.
- Partner with healthcare providers for broader data sources and real-world implementation.
- Adapt the system for use in different healthcare settings across various countries.



REFERENCES

- **UCI PIMA Indians Diabetes Dataset** – A benchmark dataset for diabetes prediction.
- **Cortes & Vapnik (1995)** – Introduced Support Vector Machine (SVM) algorithm for classification.
- **Sneha & Gangil (2019)** – Studied feature selection techniques for diabetes prediction models.
- **Ahmed et al. (2019)** – Developed a hybrid ML framework (SVM + ANN) with fuzzy logic.
- **Laila et al. (2021)** – Explored ensemble learning methods, achieving 97% accuracy.
- **Wadekar et al. (2020)** – Developed an AI-powered mobile health tool for early detection.
- **Padhy et al. (2022)** – Integrated IoT sensors with ML models, achieving 98.4% accuracy.

